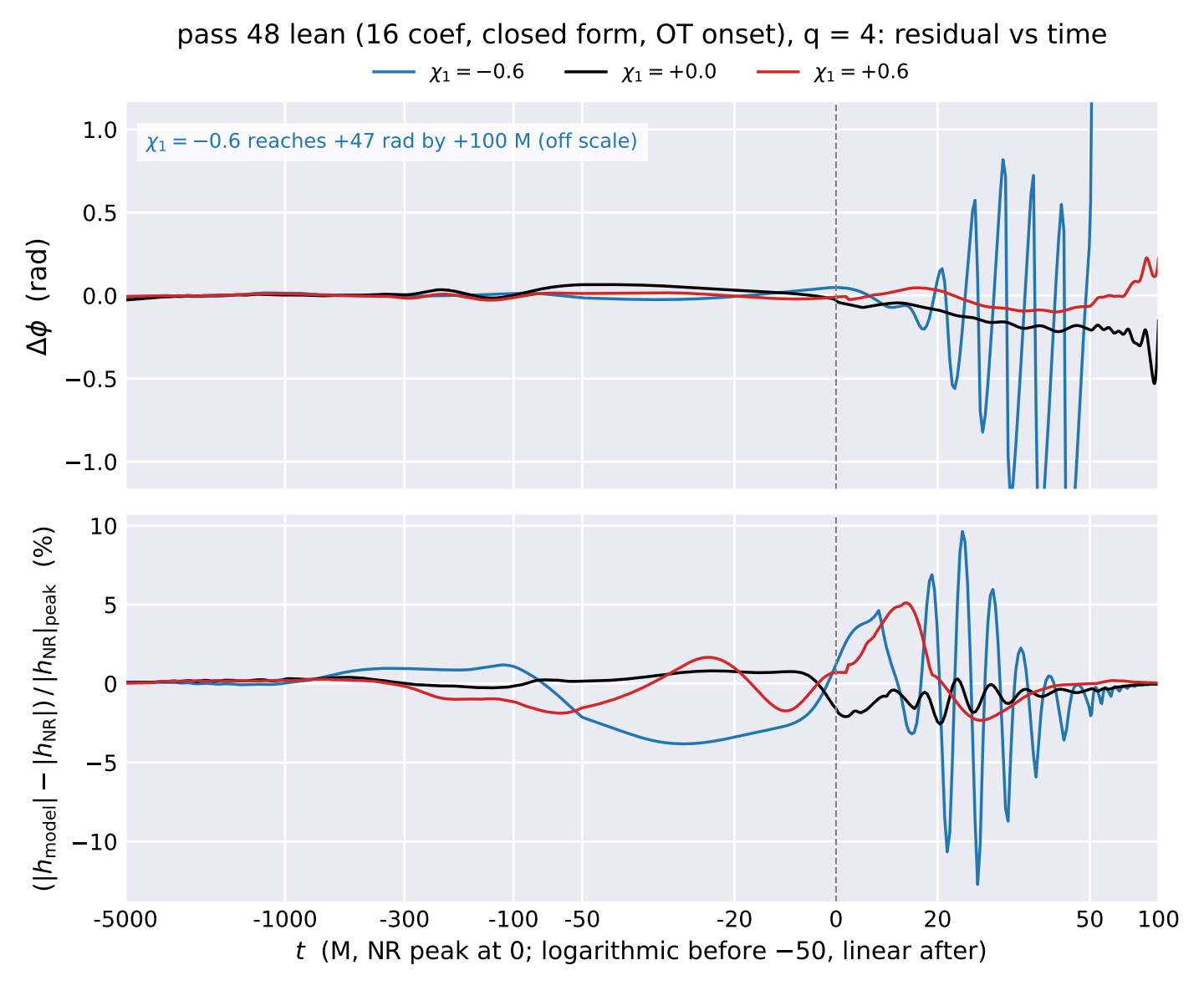
pass 48 lean (16 coef, closed form, OT onset), q = 4: residual vs time
$\chi_1 = -0.6$
$\chi_1 = +0.0$
$\chi_1 = +0.6$
$\chi_1 = -0.6$ reaches +47 rad by +100 M (off scale)
$\Delta\phi$ (rad)
1.0
0.5
0.0
−0.5
−1.0
$(|h_{\mathrm{model}}| - |h_{\mathrm{NR}}|) / |h_{\mathrm{NR}}|_{\mathrm{peak}}$ (%)
10
5
0
−5
−10
-5000
-1000
-300
-100
-50
-20
0
20
50
100
t (M, NR peak at 0; logarithmic before −50, linear after)